



Diploma in **Data Analytics**

Lesson 6: Time series analysis



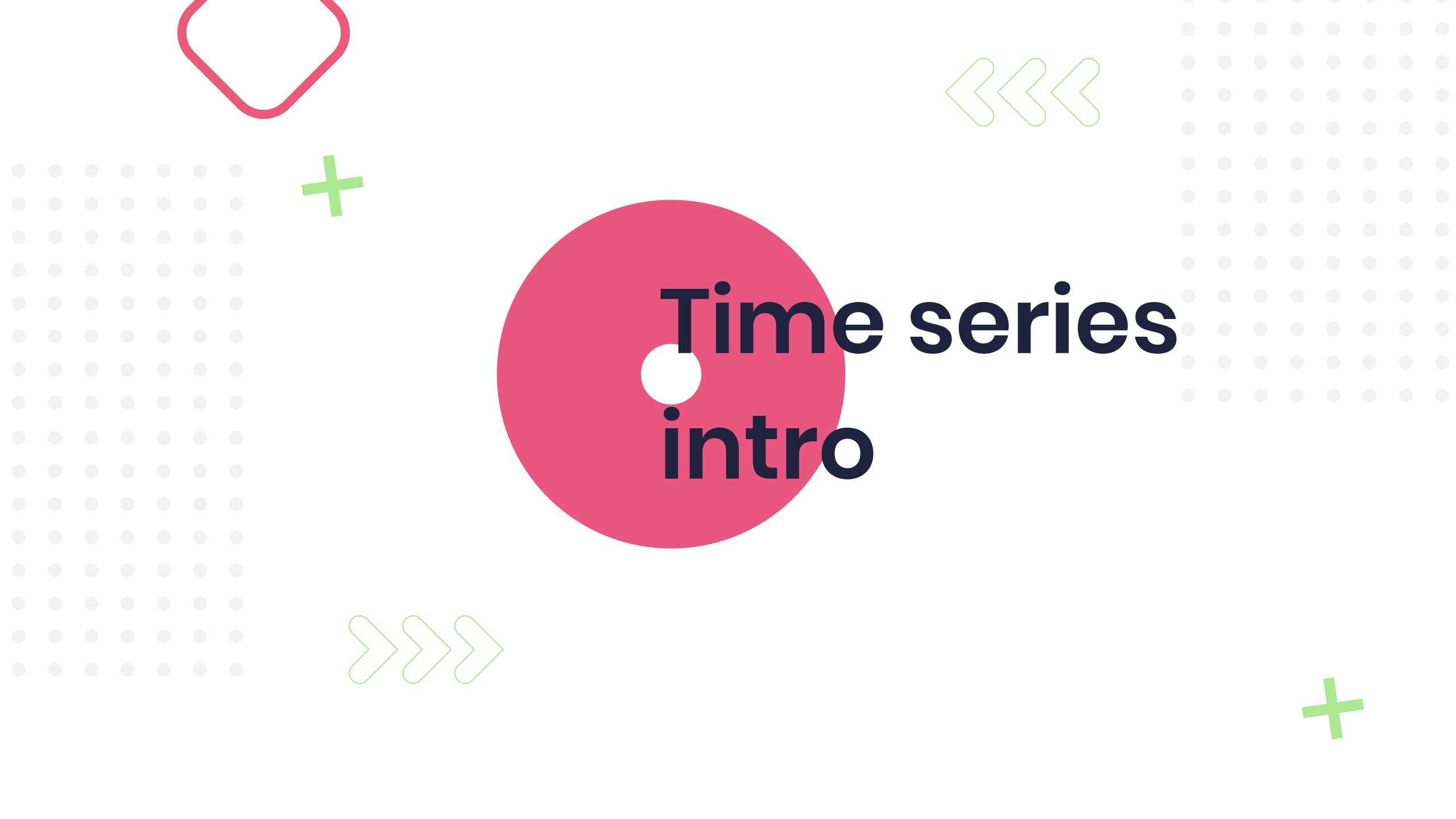
Time series intro ◀

Time series models ◀

Time series in R ◀

Lesson Objectives

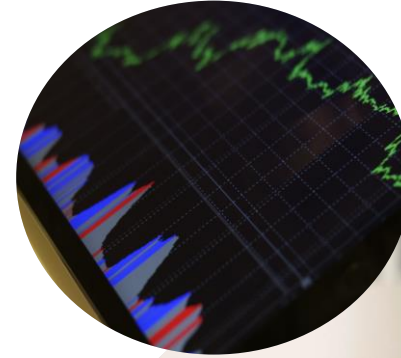
Lesson 6



Time series intro

Time series

- A variable that is measured over time in sequential order
- Analysis of time series leads to identifying trends that explain variations



Recap terminology

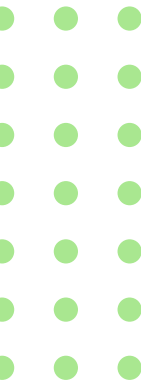
- Autocorrelation
- Seasonality
- Stationarity



Regression techniques

- Moving average
- Seasonal analysis
- Exponential smoothing
- ARIMA





**Did
you
know?**

1850 **Weather forecasting**

Robert Fitzroy created the term weather forecast and invented the tradition of printing weather forecasts in the newspaper.





Approaches to time series

- Multiplicative
 - $Y = T \times C \times S \times I$
- Additive
 - $Y = T + C + S + I$
- Combination
 - $Y = (T + C) \times S \times I$

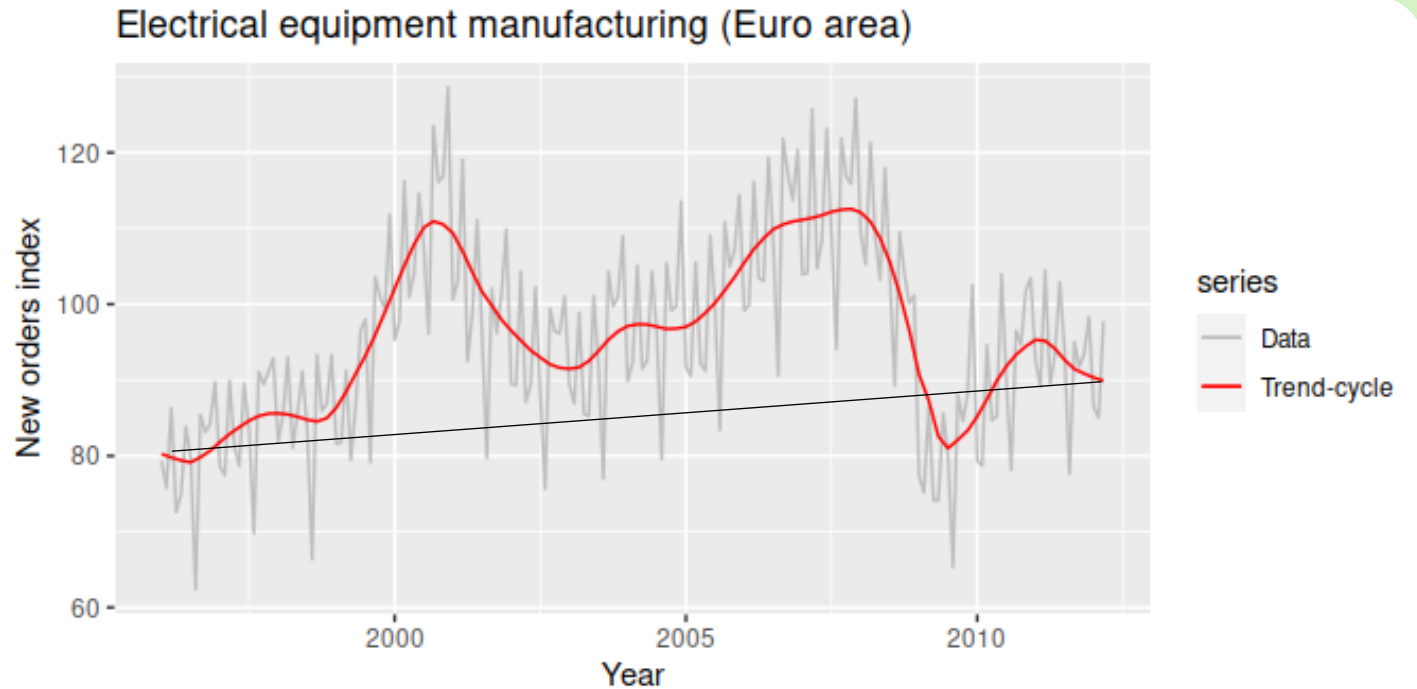
Trend variations (T)

- General tendency of data to increase or decrease over long term
- More than a year
- Ex: Population growth, technological improvements



Trend example

Electrical equipment
manufacturing over a long
term period



Source: <https://otexts.com/fpp2/components.html>

Cyclical variations (c)

- Variations that repeat over time
- More than a year from one peak to another
- Ex: Actions by government bodies inducing pessimism or optimism into economy either stimulating or depresses economic levels of activity



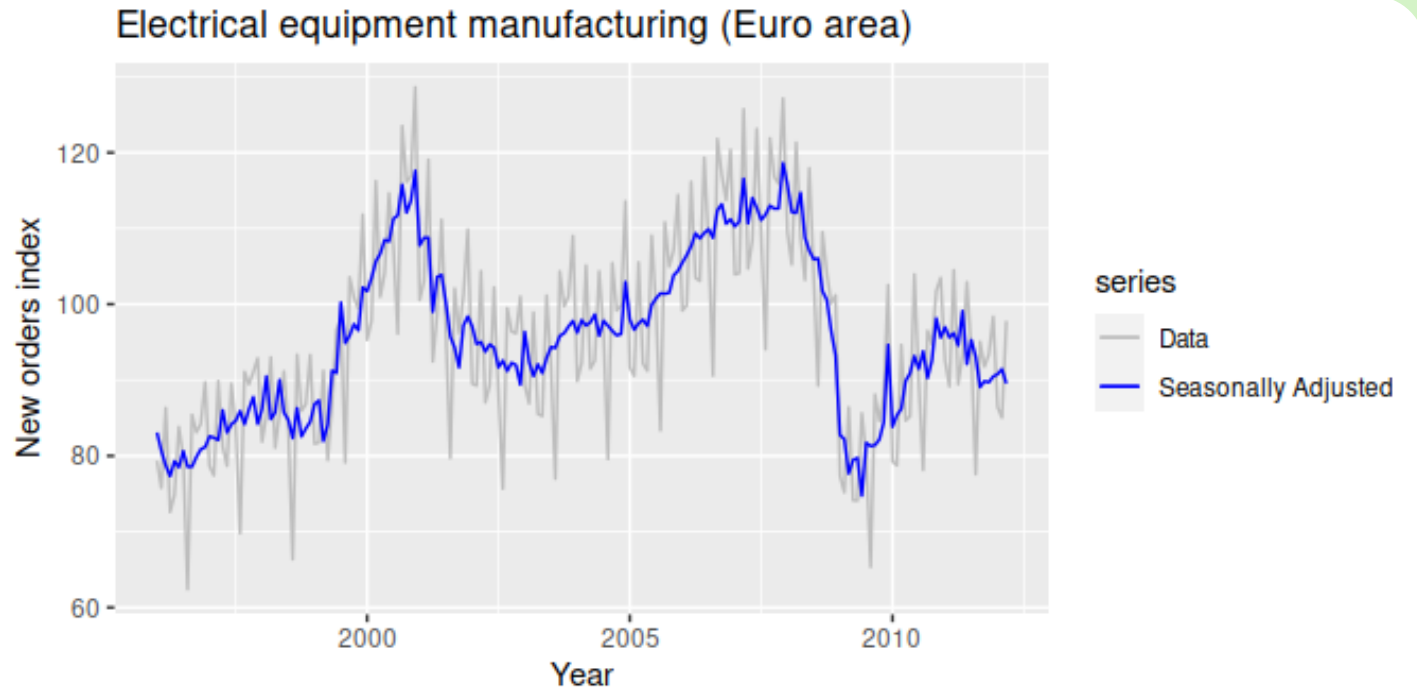
Seasonal variations (s)

- Fluctuations repeating over a period of less than a year
- Ex: Sale of heaters
- Causes: climate changes, religious events, public and school holidays



Seasonal example

Electrical equipment manufacturing with seasonal component removed from data (seasonal adjustment)



Source: <https://otexts.com/fpp2/components.html>

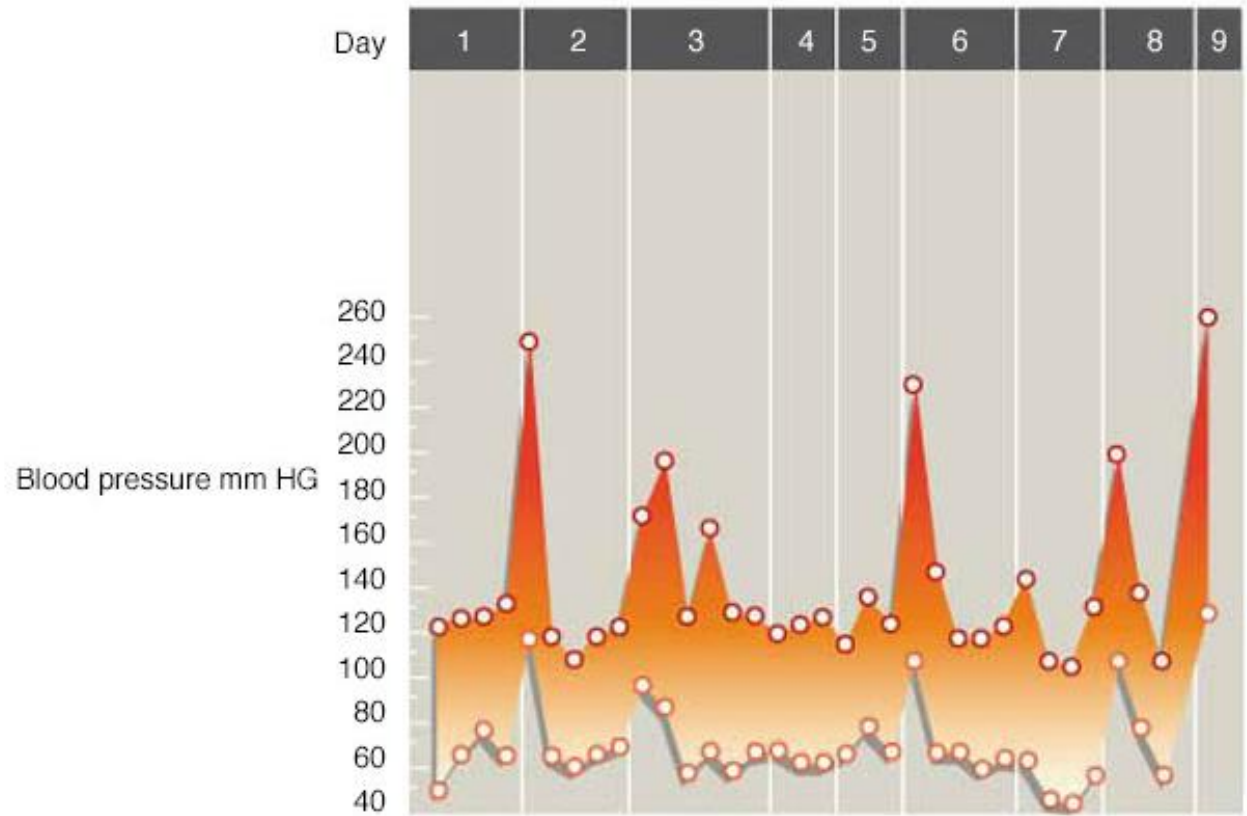
Irregular variations (I)

- Random, noise fluctuation that result from chance events
- Causes: unpredictable events like accidents, floods, strikes
- No pattern

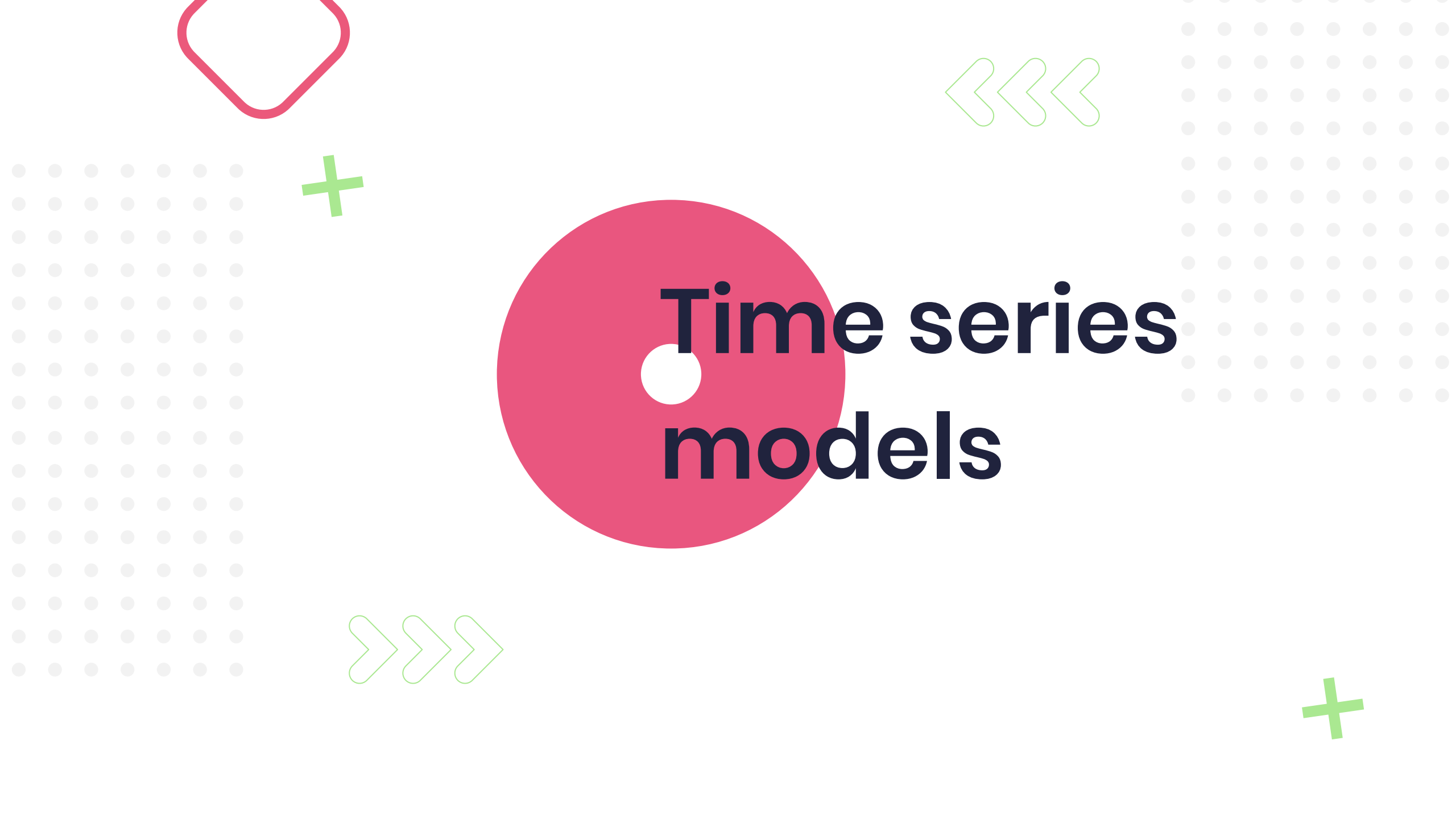


Irregular example

Pheochromocytoma can
cause random fluctuations in
blood pressure



Source: <https://www.mayoclinic.org/diseases-conditions/pheochromocytoma/symptoms-causes/syc-20355367>

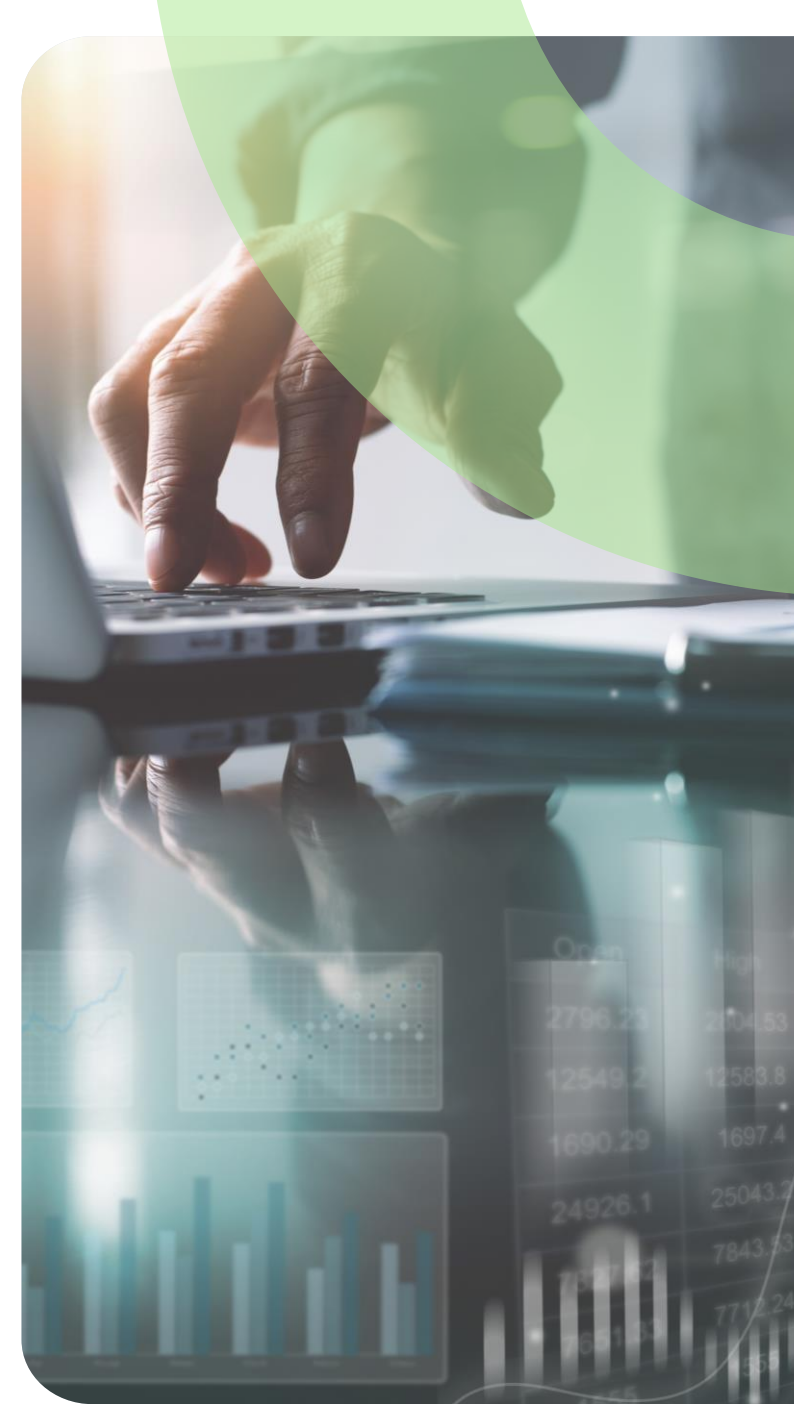


Time series models



Decomposing time series

- Linear regression techniques
 - Fit trend line
- Moving average method



Regression techniques

- Fit a trend line to the data
 - Linear
 - $\hat{y} = a + bx_i$
 - Quadratic
 - $\hat{y} = a + bx_i + cx_i^2$



Moving average method

- Take away short-term seasonal and irregular variations in order to leave behind combined trend and cyclical components
- Smooths out short-term fluctuations



Moving average

Moving average

$$= \frac{\Sigma(\text{most recent } k \text{ data values})}{k}$$

,where k = period number





Moving average example

Monthly sales of a company
from January to December
2010

2010	Sales	2010	Sales
January	7	July	12
February	3	August	4
March	5	September	10
April	9	October	13
May	7	November	9
June	9	December	10

Source: Introductory Statistics, Department of Statistics and Population studies of the University of the Western Cape





Moving average example

First three-period moving
average

2010	Sales
January	7
February	3
March	5
Average	5

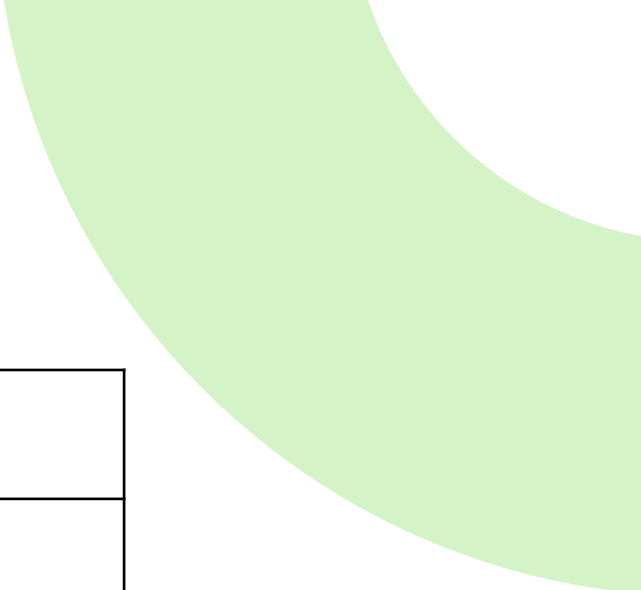




Moving average example

Second three-period moving
average

2010	Sales
February	3
March	5
April	9
Average	5,6667





Moving average example

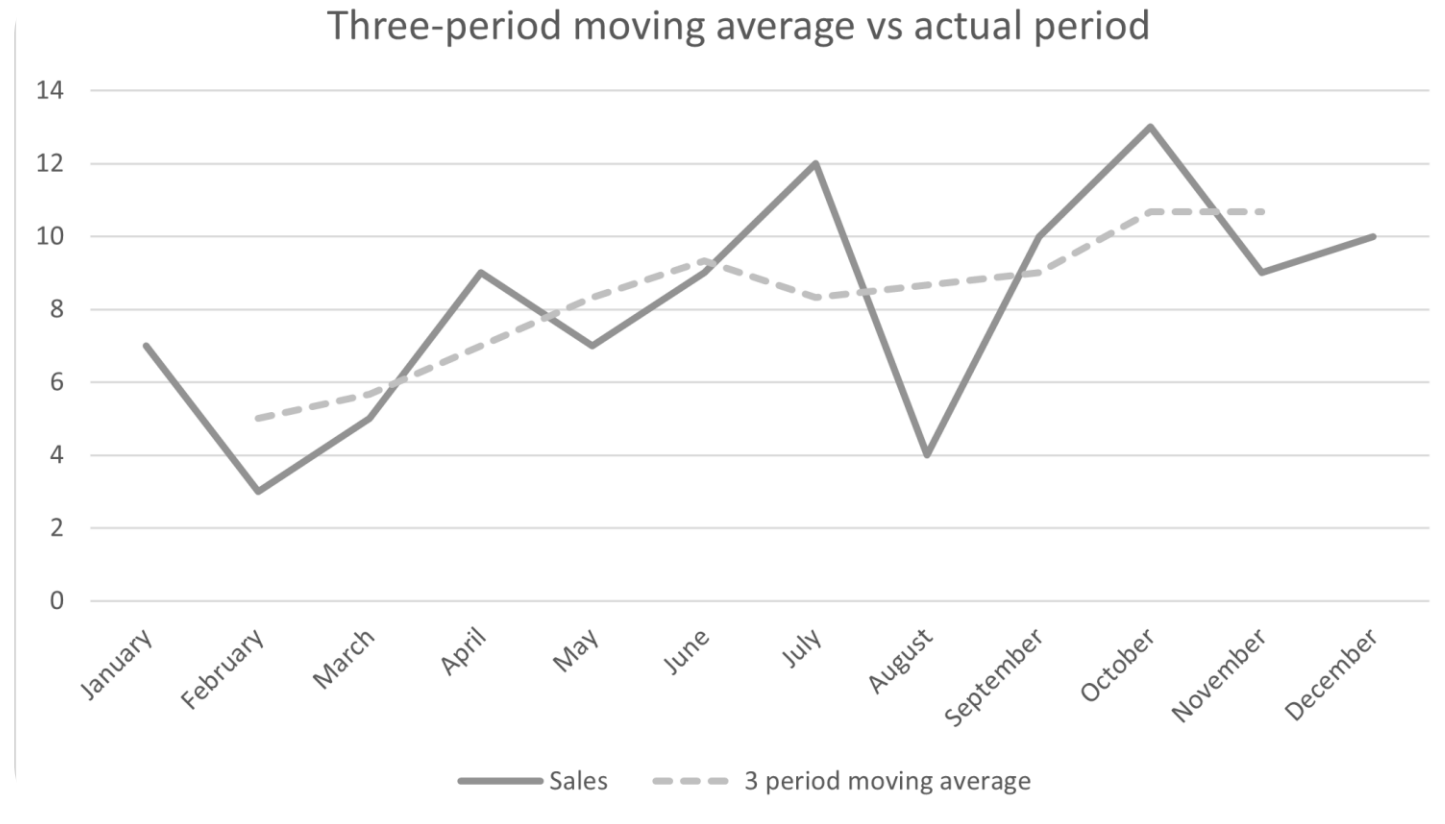
Third three-period moving
average

2010	Sales
March	5
April	9
May	7
Average	7



Moving average example

Three-month moving average
compared to original data



Seasonal analysis

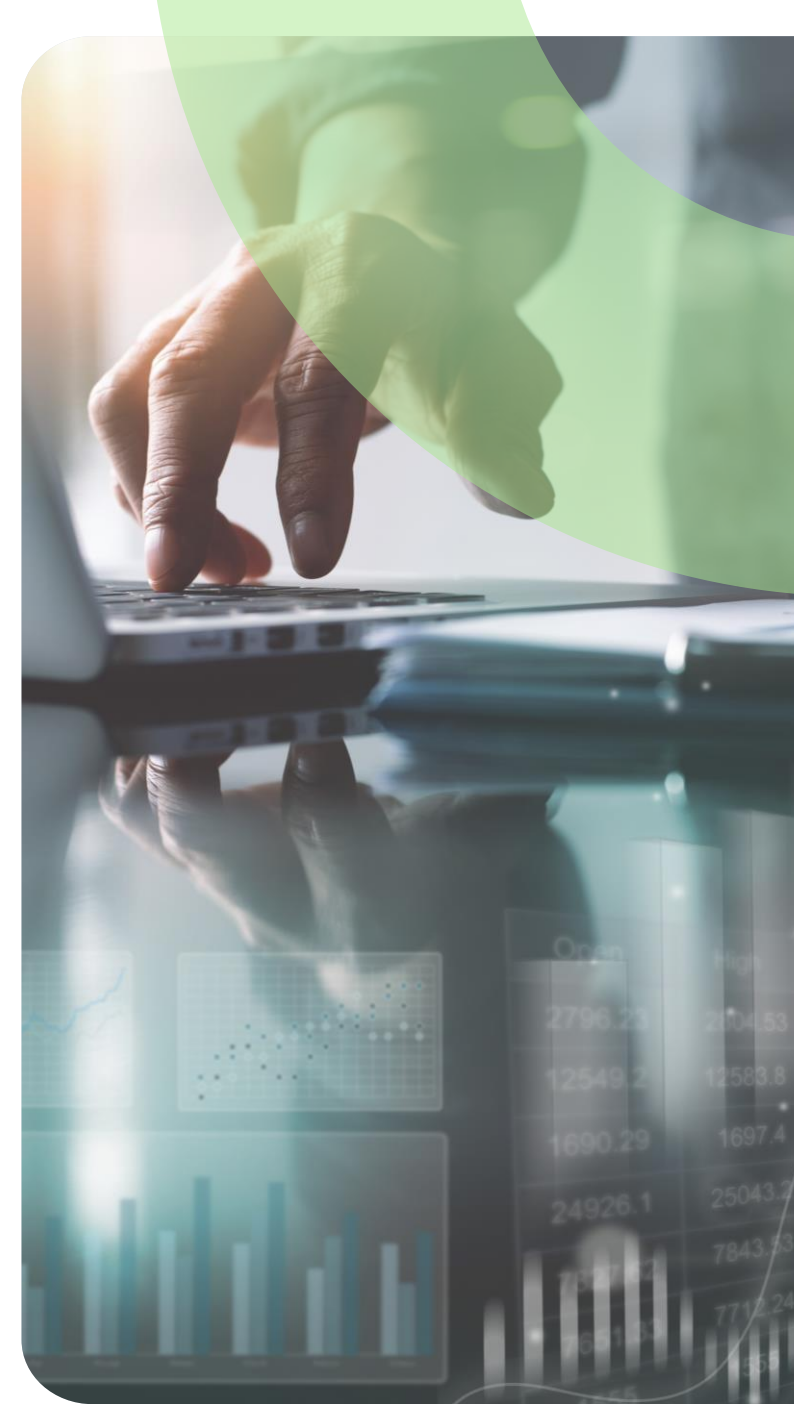
- Seasonal patterns that occur at regular interval less than a year apart
- Seasonal index factor can help compensate for the effect of a pattern





Deseasonalising time series

- Remove seasonal influence
- Able to only account for influence of trend-, cyclical- and irregular variations





Challenge >>

Apply seasonal adjustment to time series:

<http://robjhyndman.com/tsdldata/data/nybirths.dat>

#exploredata